

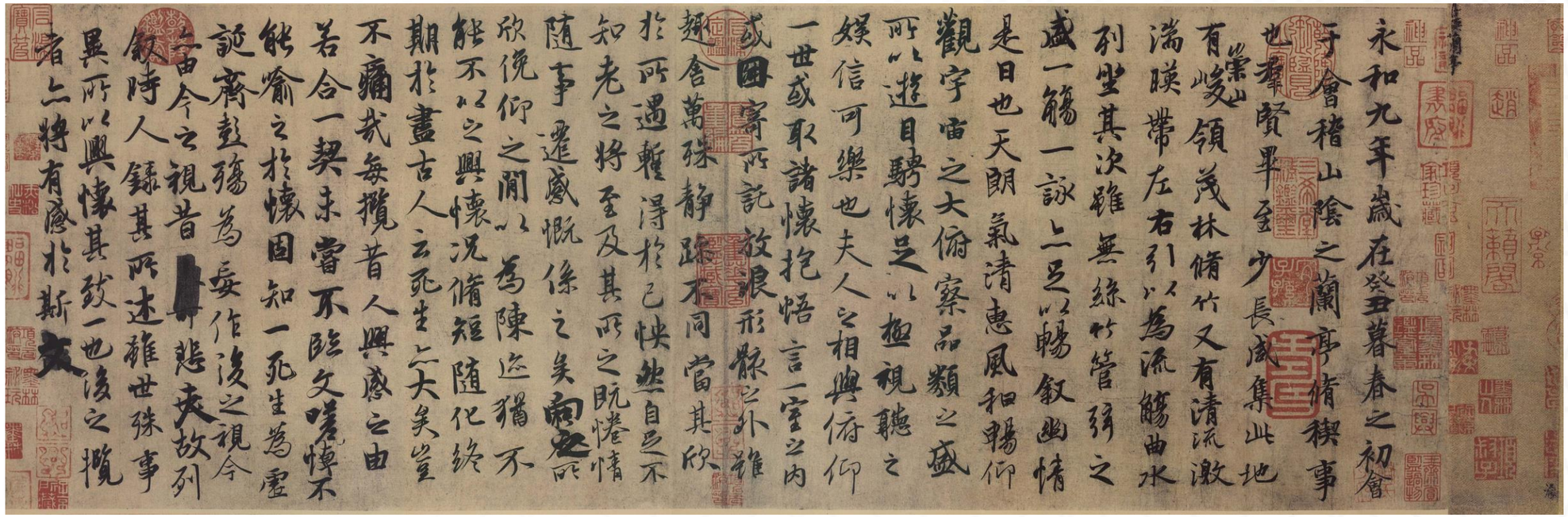
Calli-VA: A Visual Analytics System for Analyzing and Comparing Chinese Calligraphic Styles

Jincheng Li[1,2], Jinpeng Wu[1], Shaocong Tan[2], Lin Du[1], Yu Zhang[3], Chaofan Yang[2], Jiadi Zhang[2], Rebecca Ruige Xu[4], Rui Shi[2], Lu Bai[1], and Xiaoru Yuan[2]

[1] Beijing Normal University [2] Peking University [3] University of Oxford [4] Syracuse University

Chinese Calligraphy

- A quintessential element of Chinese cultural heritage
  - Blend the functional purpose of writing with profound aesthetic expression
- Numerous renowned calligraphers and notable works have emerged





Lantingji Xu
“The world’s No.1 semi-cursive script”
— Wang Xizhi

Calligraphy Style Analysis

- Difference in calligraphic styles
  - Influence between calligraphers
  - Evolution within the works of the same calligrapher at different stages
- Analyzing and comparing styles can
  - enhance the appreciation, learning, and advancement of calligraphy
  - gain valuable insights into ancient China

Wang Xizhi



Su Lingzhi



Variation between calligraphers

“More thick and compact.”

Yizhou Tiexiang Song



Meng Zhenrong Bei

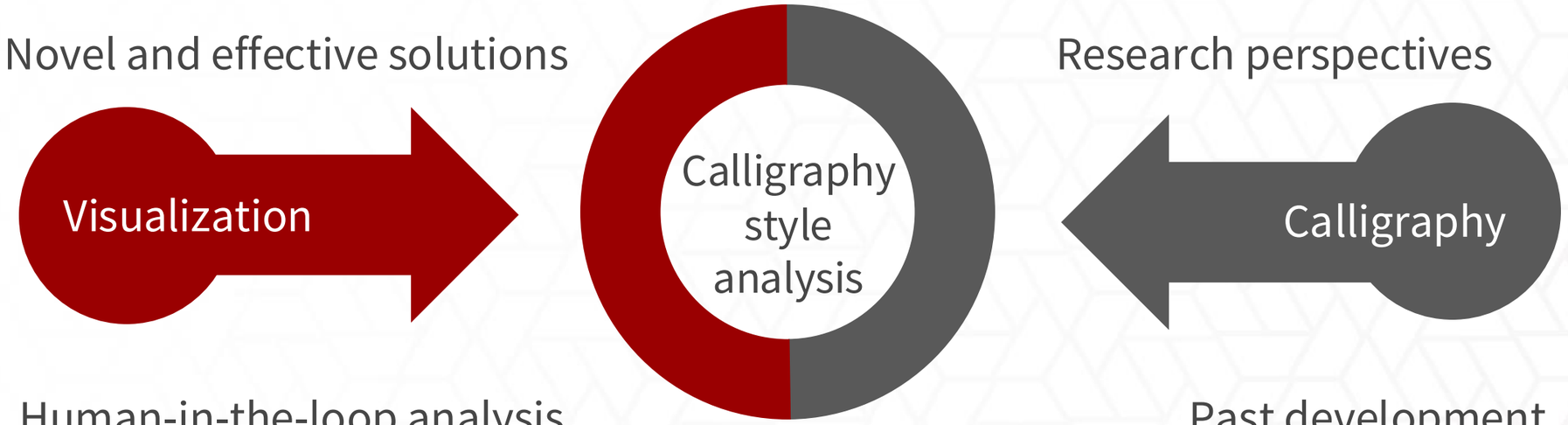


Variation within the works of Su Lingzhi

“Less thick, more dynamic.”

Interdisciplinary Collaboration

Goal: provide calligraphy scholars with full-process support



Human-in-the-loop analysis

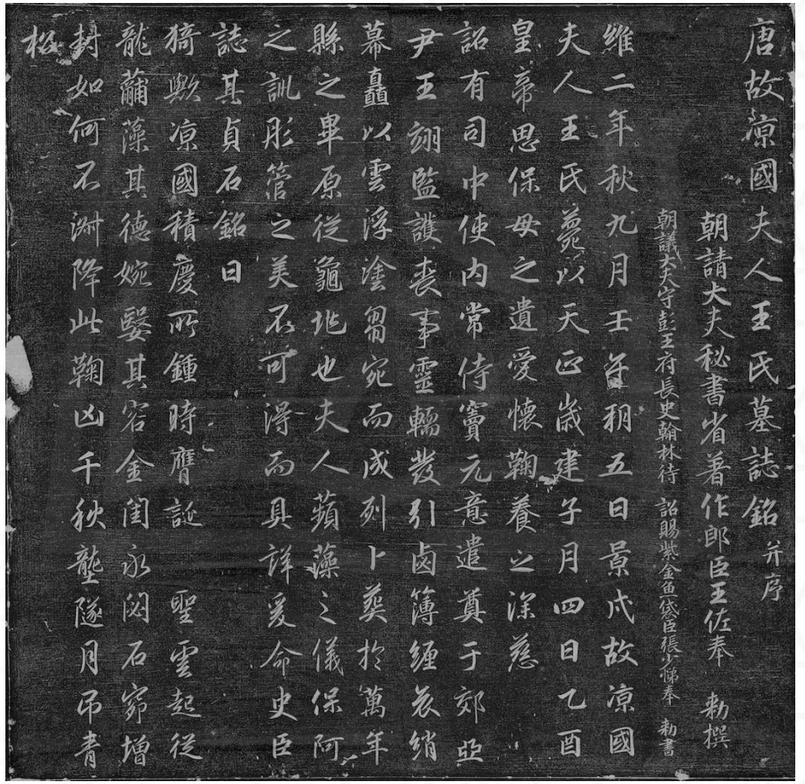
Intuitive data presentation
Interactive and iterative analysis

Past development

Unique domain data
Complex analytical problems

Data Description

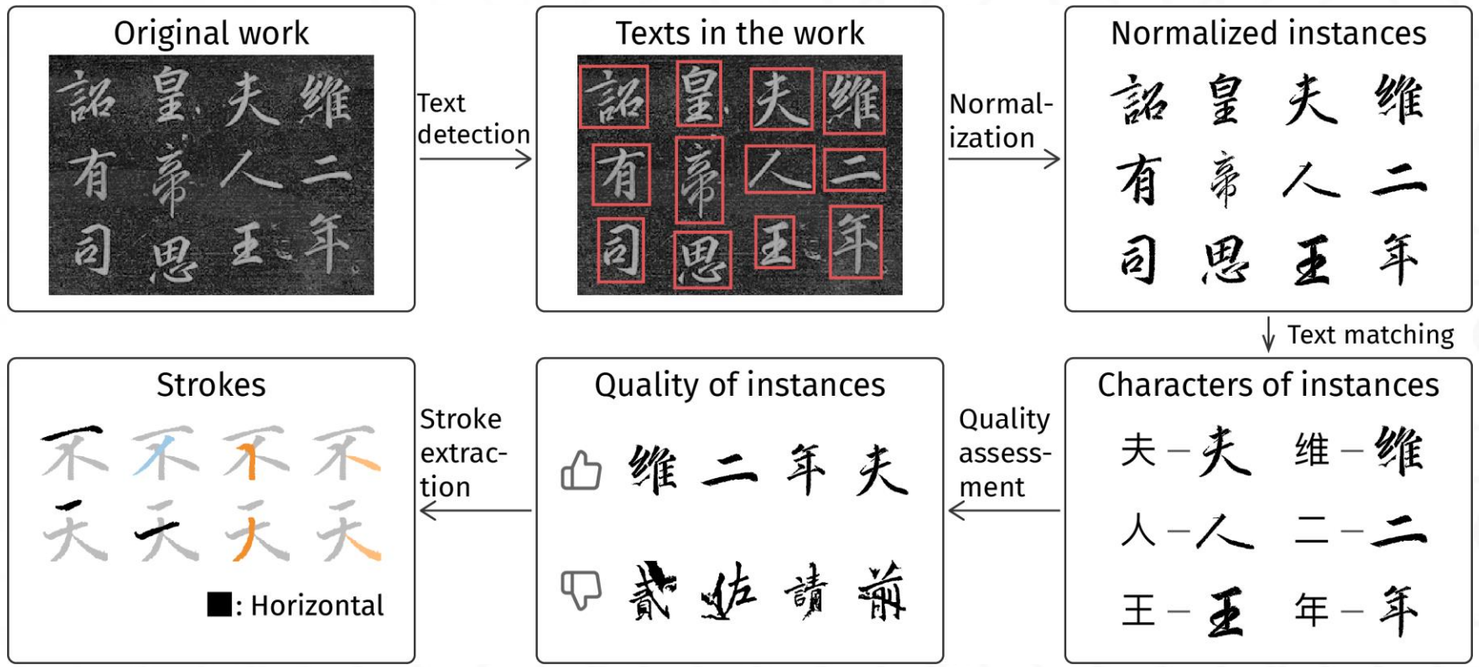
- 17 inscriptions by 12 calligraphers
  - Content
    - 17,835 character instances
    - Metadata: text, images, and temporal information
  - All related to Wang Xizhi
    - Calligraphers heavily influenced by Wang's style (15)
    - Characters sourced from Wang's original calligraphy (2)


唐故涼國夫人王氏墓誌銘并序
朝請大夫秘書省著作郎臣王佐奉 勅撰
朝議大夫守彭王府長史翰林待 詔賜紫金魚袋臣張少悌奉 勅書
維二年秋九月壬子朔五日景戌故涼國
夫人王氏薨以天亓歲建子月四日乙酉
皇帝思保母之遺愛懷鞠養之深慈
詔有司中使内常侍竇元意遣冀于郊收
尹王翊監護喪事靈輀發引鹵簿鉦笳繡
幕龜以雲浮塗罔宛而成列卜葬於萬年
縣之卑原從龜北也夫人薤藻之儀保阿
之訓彤管之美不可湮而具詳爰命史臣
誌其貞石銘曰
猗歟涼國積慶所鍾時膺誕
龍蘭藻其德婉嬰其容金閨永閟石窌增
封如何不淑降此鞠凶千秋壟隧月吊書
柩

Suzong Baomu Wang Shi Muzhi Bing Gai
— Zhang Shaoti

Data Processing — Character and Stroke Extraction

- Extract instance images and match them to text
- Filter out low-quality instances
- Extract strokes to capture finer stroke and local structural details



Data Processing — Character Characterization

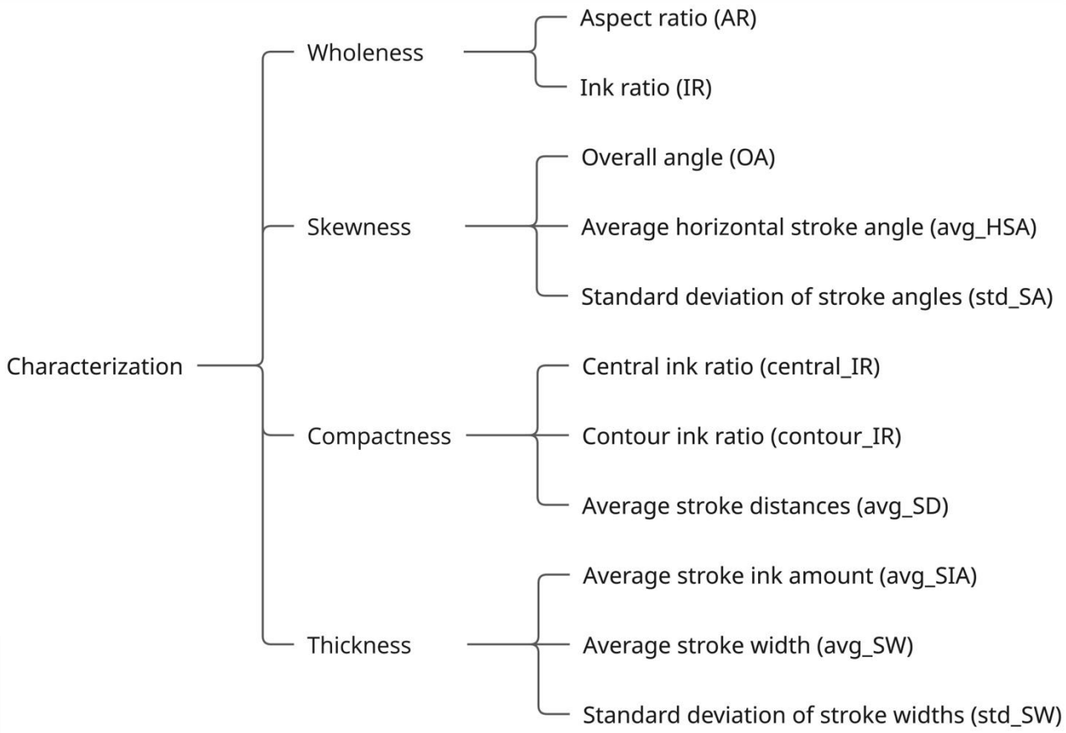
- Based on “Pinghua Kuanjie” and “Xiehua Jinjie” [Sha, 1987]
  - “Pinghua Kuanjie”: relatively flat horizontal strokes and ample spacing
  - “Xiehua Jinjie”: more slanted horizontals, denser spacing, minimal negative space
- Reflect the perspectives of wholeness, skewness, compactness, and thickness



Pinghua Kuanjie



Xiehua Jinjie



Design Considerations

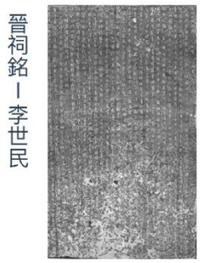


Before analysis

Select characters to analyze

Entry
- Provide an overview of works and calligraphers
- Provide an overview of characters

Overview — Work Overview

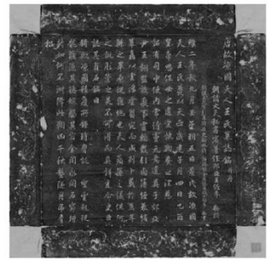

晉祠銘 — 李世民

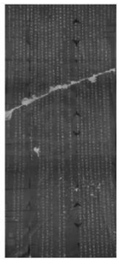

劉德墓誌 — 佚名1


高福墓誌 — 佚名2


易州鐵像碑頌 — 蘇靈芝


靈運塔銘 — 勤口

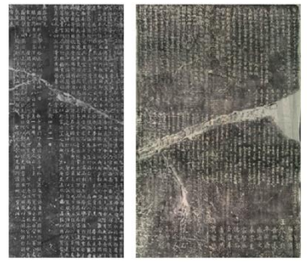

張少悌 肅宗保母王氏墓誌并蓋 —

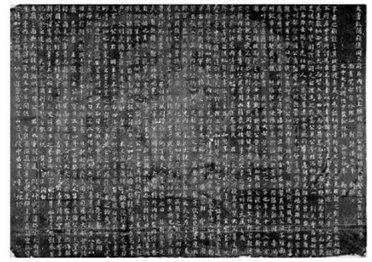

序記并心經 — 王羲之 懷仁集王羲之書三藏聖教

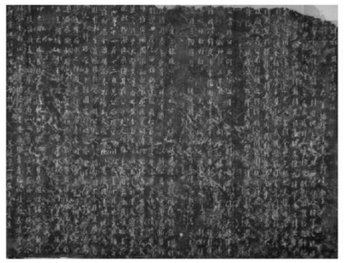

李勣碑 — 王持


昇仙太子廟碑 — 武曌


李思訓神道碑 — 李邕


夢真容碑 — 蘇靈芝


高力士墓誌 — 張少悌


興福寺碑 — 王羲之


隆闡大法師碑 — 懷惲


麓山寺碑 — 李邕


田琬德政碑 — 蘇靈芝

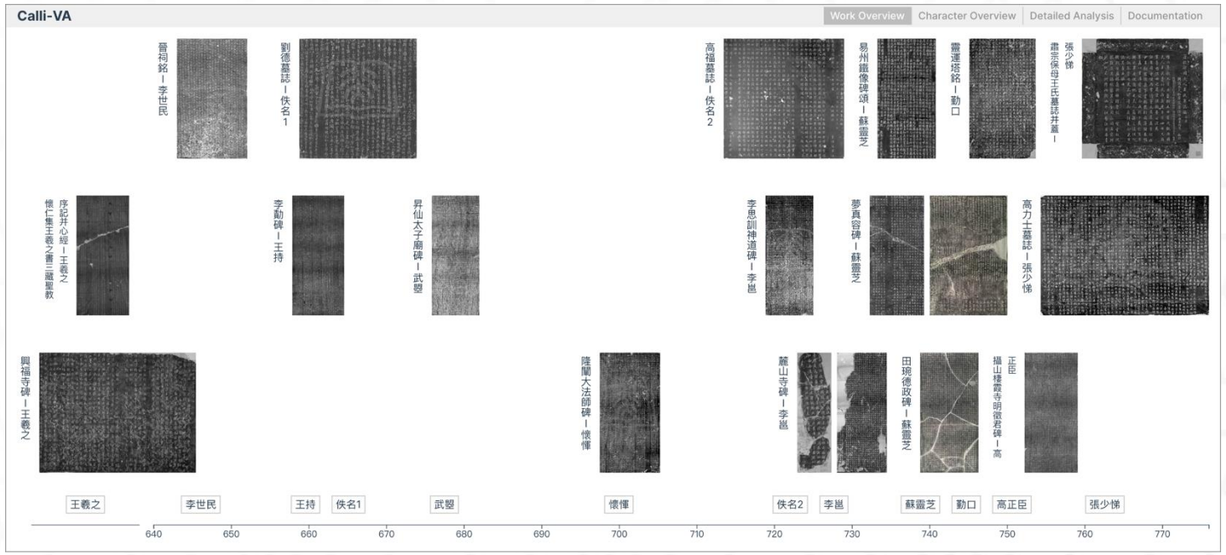

正臣 攝山棲霞寺明徵君碑 — 高

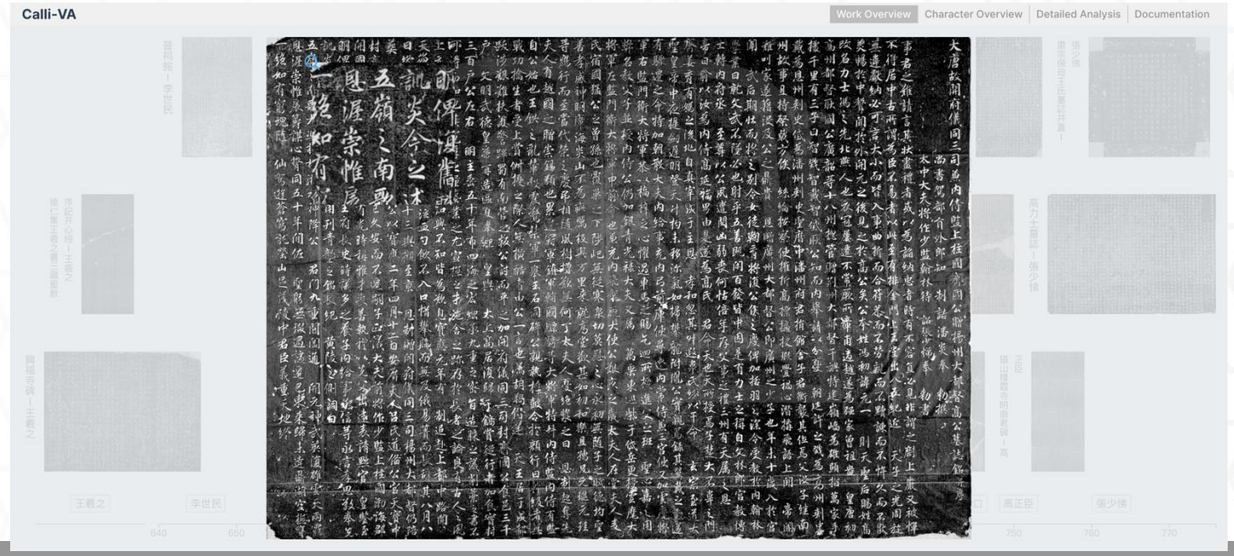
王羲之 | 李世民 | 王持 | 佚名1 | 武曌 | 懷惲 | 佚名2 | 李邕 | 蘇靈芝 | 勤口 | 高正臣 | 張少悌

640 650 660 670 680 690 700 710 720 730 740 750 760 770

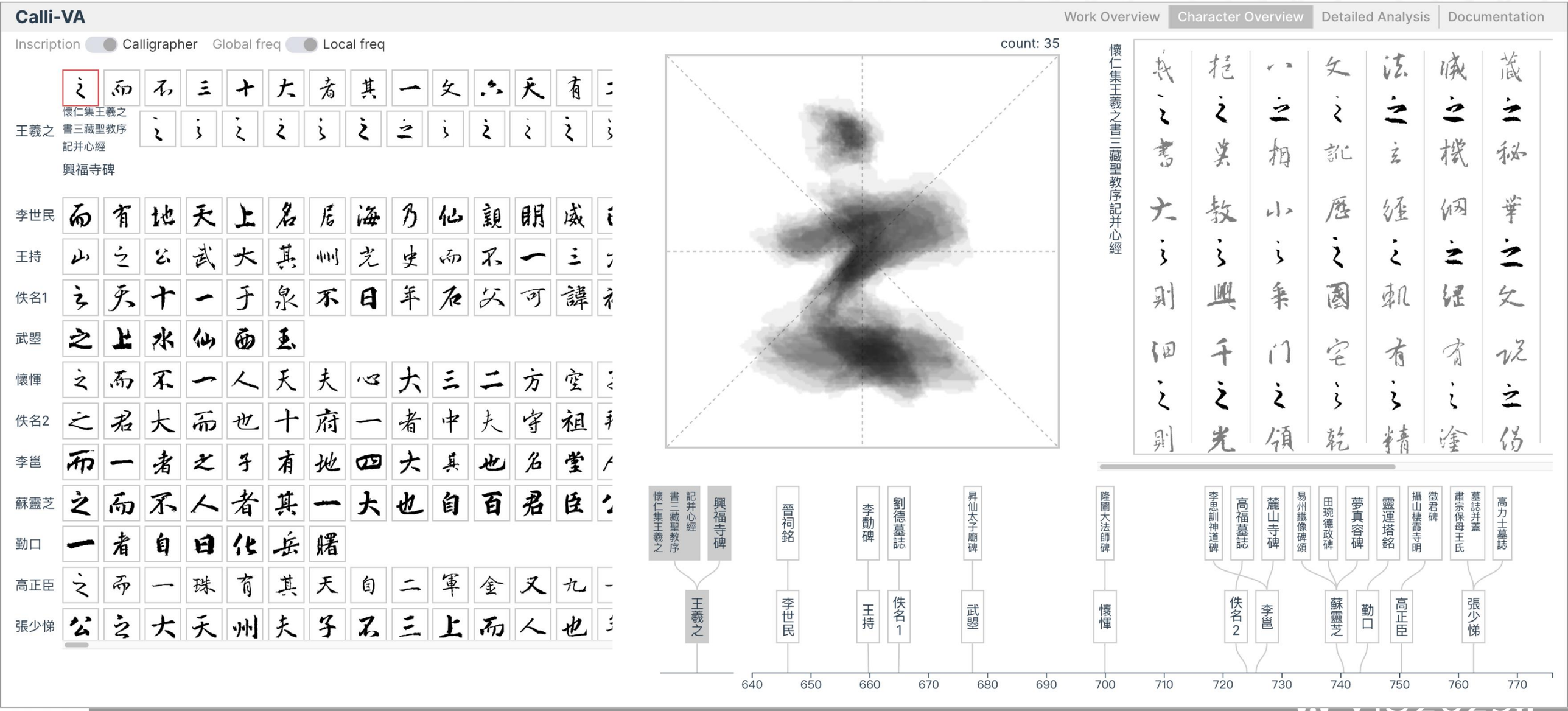
Overview — Work Overview

- Display inscriptions with corresponding calligraphers
  - X-axis: mainly arranged by time
    - Two inscriptions by Wang Xizhi are placed at the leftmost
    - Fine-tuned to resolve overlaps
- Click an inscription to view details





Overview — Character Overview

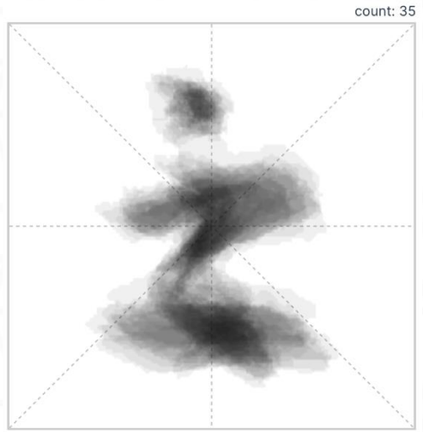


Overview — Character Overview

- Explore characters
- Character Matrix View: organize characters into a matrix
- After clicking a character
  - Superimposed Character View: overlay instances
  - Character Context View: display each instance's neighboring characters
- Timeline View: show correspondence between inscriptions and calligraphers



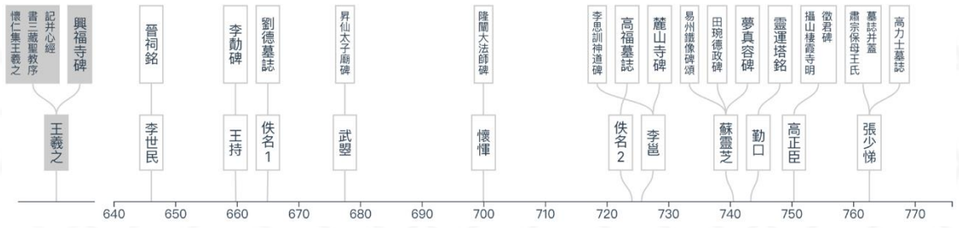
Character Matrix View



Superimposed Character View



Character Context View



Timeline View

VIS2025

Design Considerations



Before analysis

Select characters to analyze



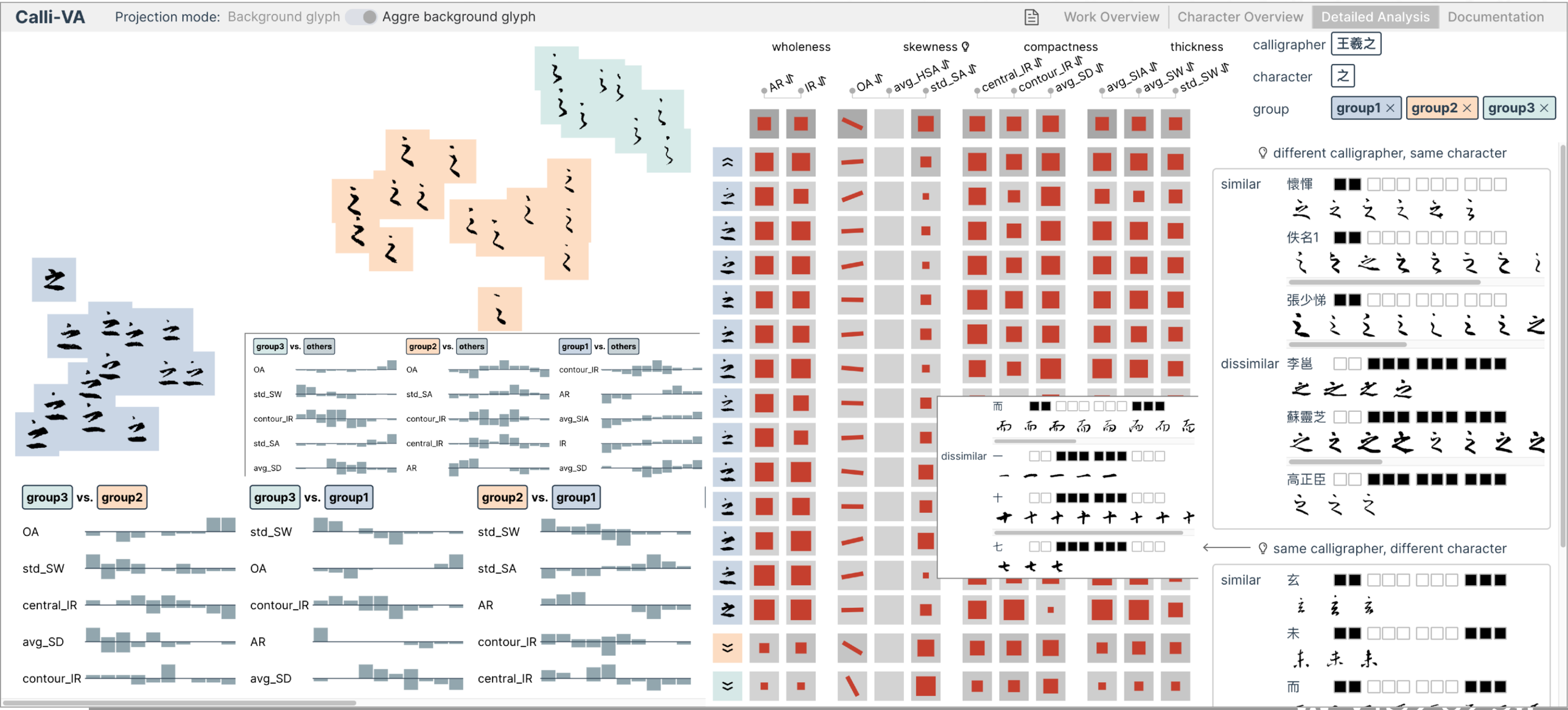


During analysis

Support the various analysis scopes

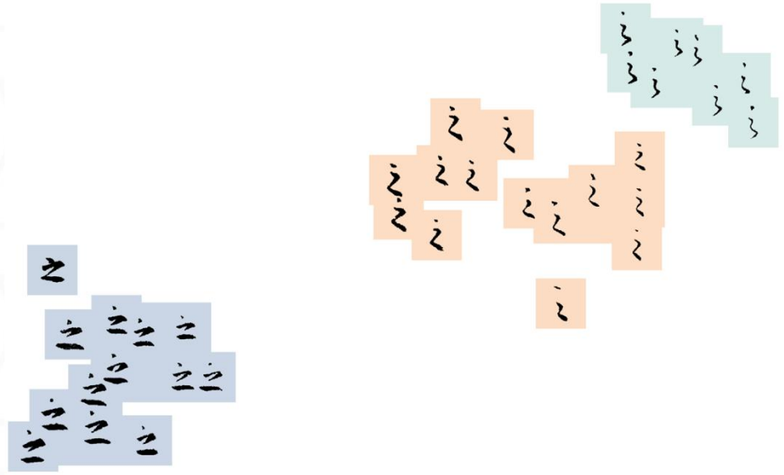
- Entry
  - Provide an overview of works and calligraphers
  - Provide an overview of characters
- Exploration
  - Explore characters from multiple perspectives
  - Explore characters across multiple granularities
- Explanation → Integrate pattern explanation mechanisms
- Recommendation → Facilitate adjustments to granularities and perspectives

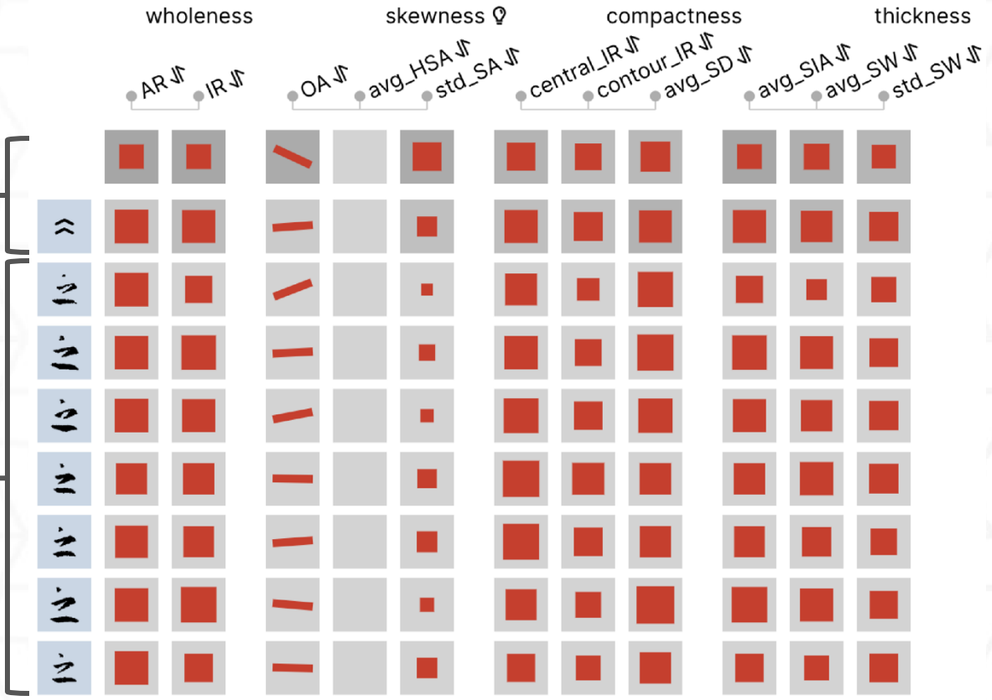
Detailed Analysis



Detailed Analysis — Exploration

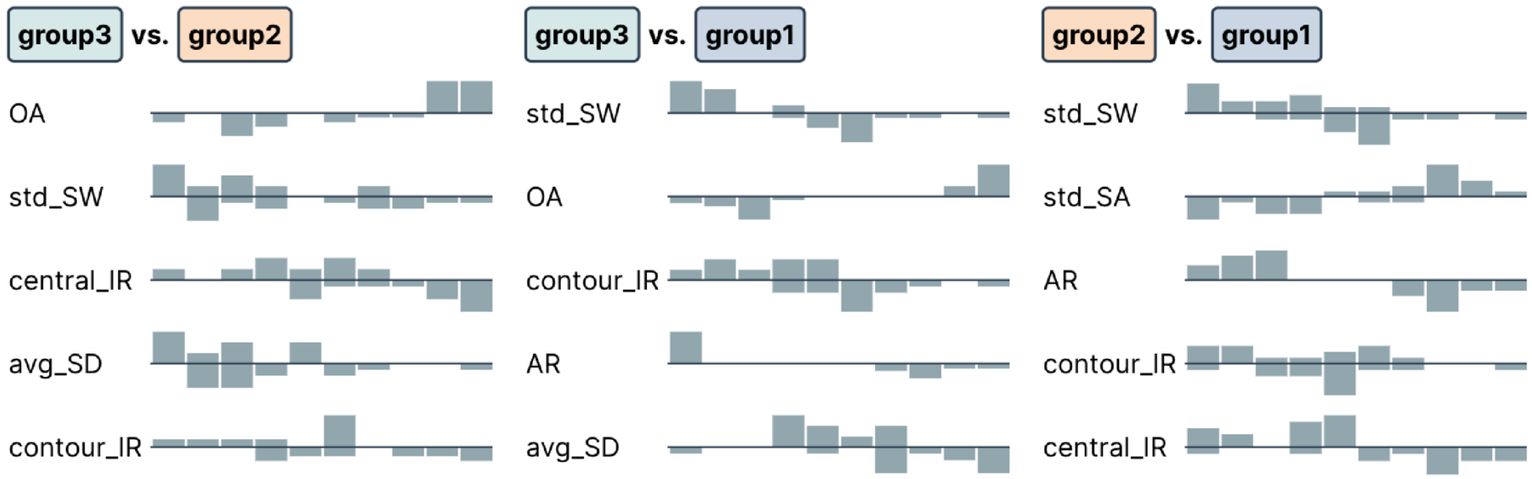
- Analyze the selected character instances
- Projection View: show relationships among instances
  - Enable to group instances using lasso brushing
- Table View: present detailed feature values
  - Row types
    - Aggregated rows: summaries for subsets
    - Instance rows: data for individual instances





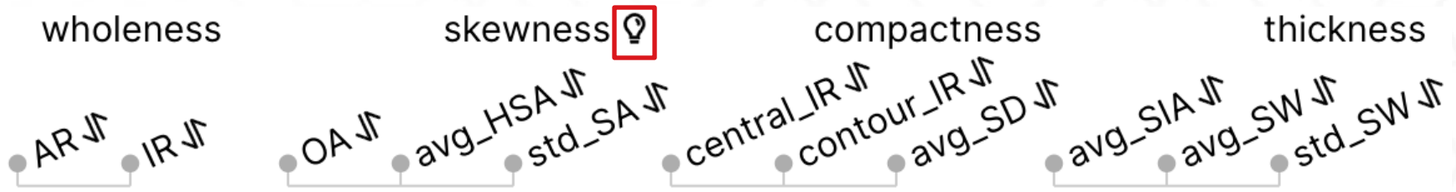
Detailed Analysis — Explanation

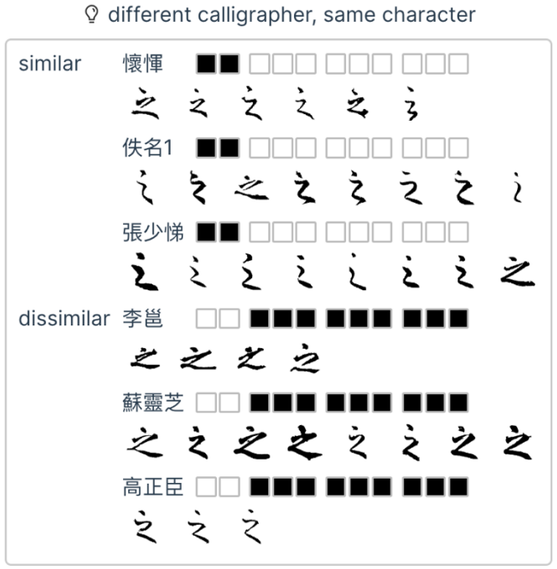
- Pattern Explanation View: interpret patterns revealed in the Projection View
  - Identify the contribution of features by ccPCA [Fujiwara et al., TVCG 2020]
  - Show distributions of the top five features as histograms

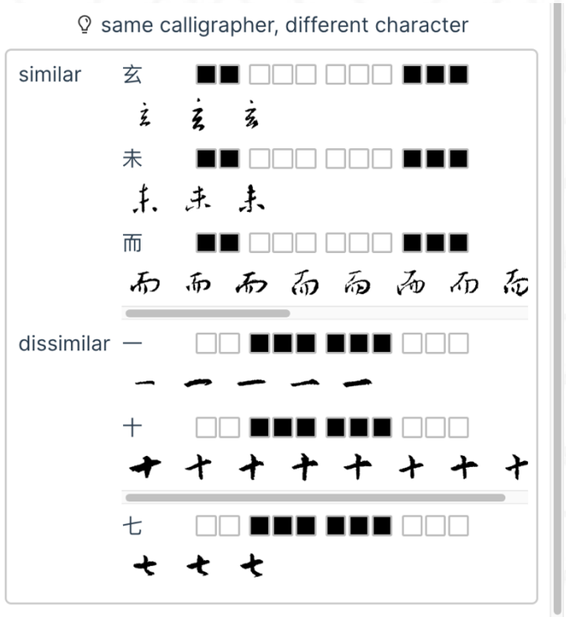


Detailed Analysis — Recommendation

- Adjust analysis granularities and perspectives
- Granularity Recommendation View
  - Two categories
    - Same characters across different objects (calligraphers or works)
    - Different characters within the same objects
  - Two recommendation types: similar and dissimilar characters
- Recommend perspectives where instances form clusters







Design Considerations



Before analysis

Select characters to analyze

- Entry
  - Provide an overview of works and calligraphers
  - Provide an overview of characters



During analysis

Support the various analysis scopes

- Exploration
  - Explore characters from multiple perspectives
  - Explore characters across multiple granularities
- Explanation → Integrate pattern explanation mechanisms
- Recommendation → Facilitate adjustments to granularities and perspectives



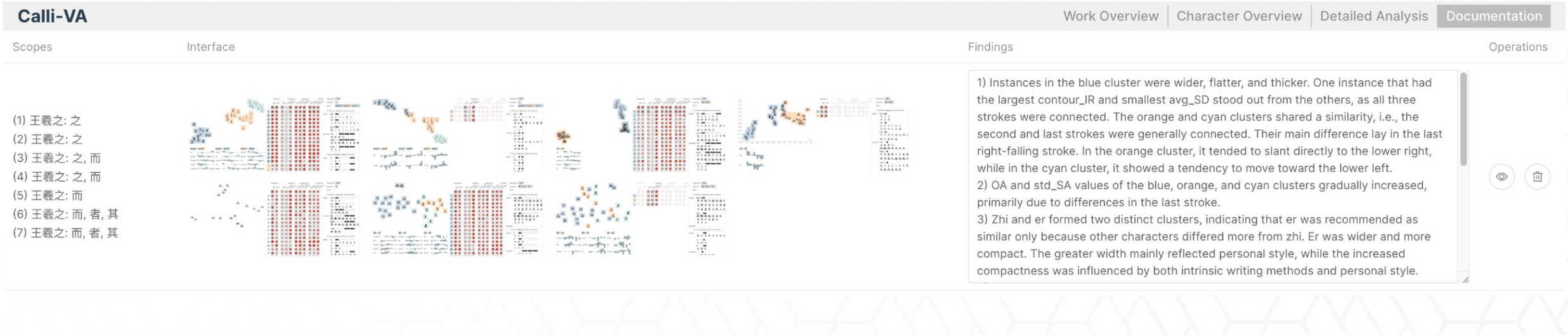
After analysis

Record findings

- Documentation
  - Allow saving intermediate results
  - Provide necessary auxiliary information

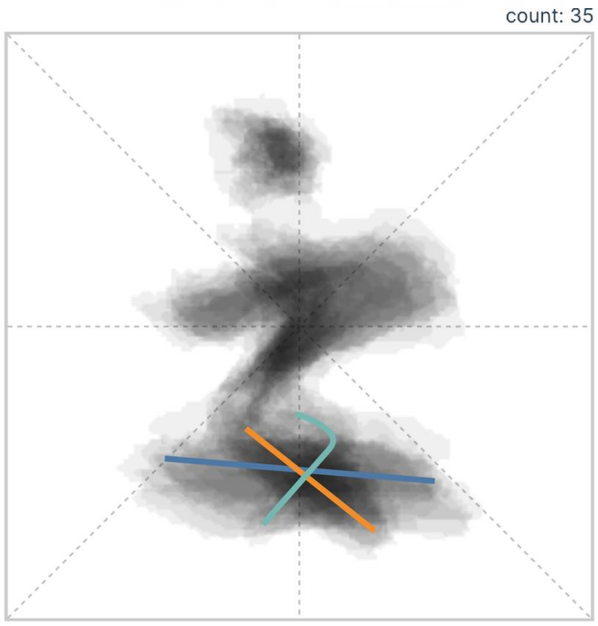
Documentation

- Summarize findings
- Table-based layout
  - Row: a round of analysis
  - Record findings by text

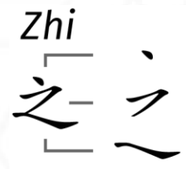


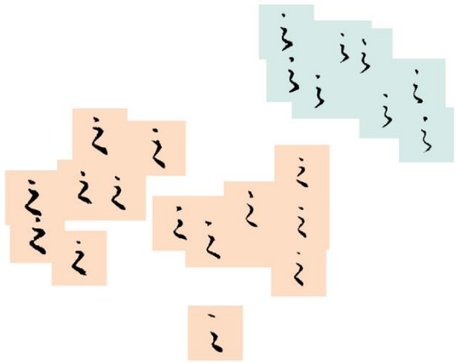
Case — Analyzing the Character *Zhi*

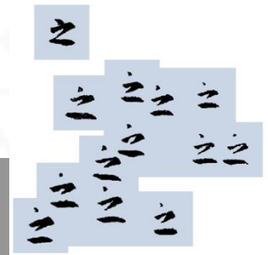
- Start from Wang Xizhi
  - Superimposed Character View: reveal three main styles
    - Differ primarily in the last right falling stroke
  - Character Context View: show variation in the second left horizontal falling stroke
    - Whether it connects to the last stroke
  - Projection View: form three distinct clusters



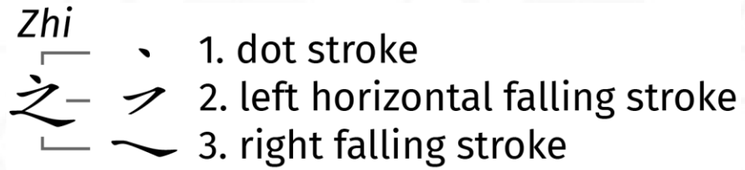




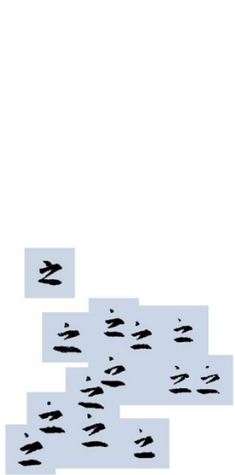


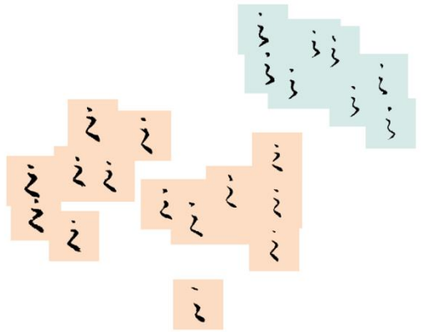


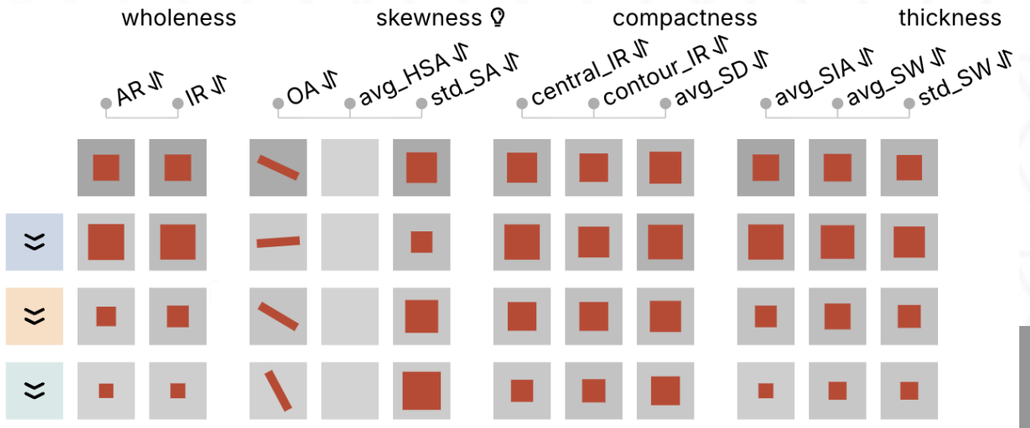
Case — Analyzing the Character *Zhi*

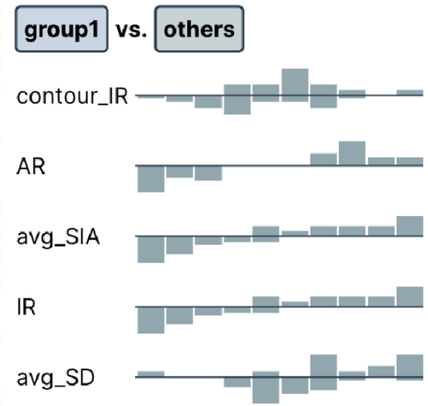


- Start from Wang Xizhi
  - Table View and Pattern Explanation View reveal:
    - Blue cluster: wider (↑AR), flatter (↓OA), and thicker (↑IR, ↑avg_SIA)
    - Orange and cyan clusters
      - Similarity: both connect the second and last strokes (↓avg_SD)
      - Difference: last stroke — orange leans lower right (↓OA), cyan leans further lower left (↑OA)
  - ⇒ Wang Xizhi developed diversified style
  - ⇒ “Every stroke must contain a certain degree of variation”





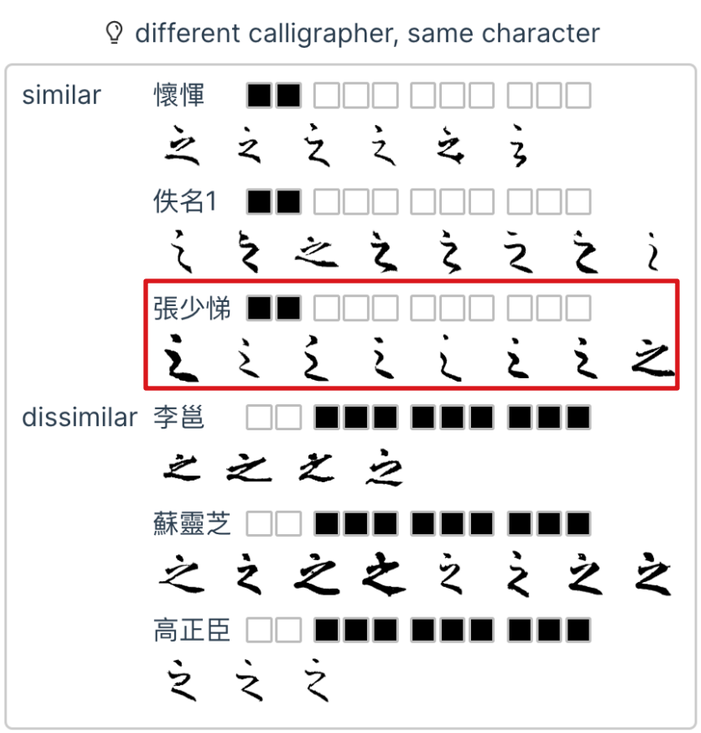


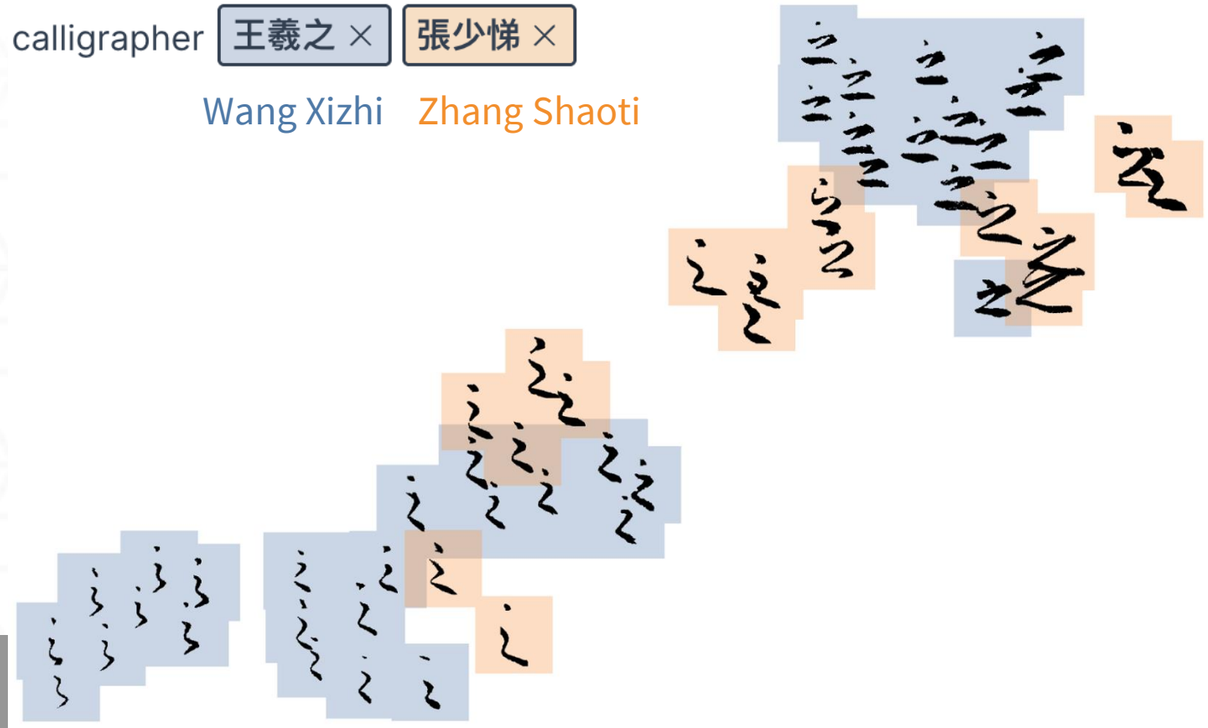




Case — Analyzing the Character *Zhi*

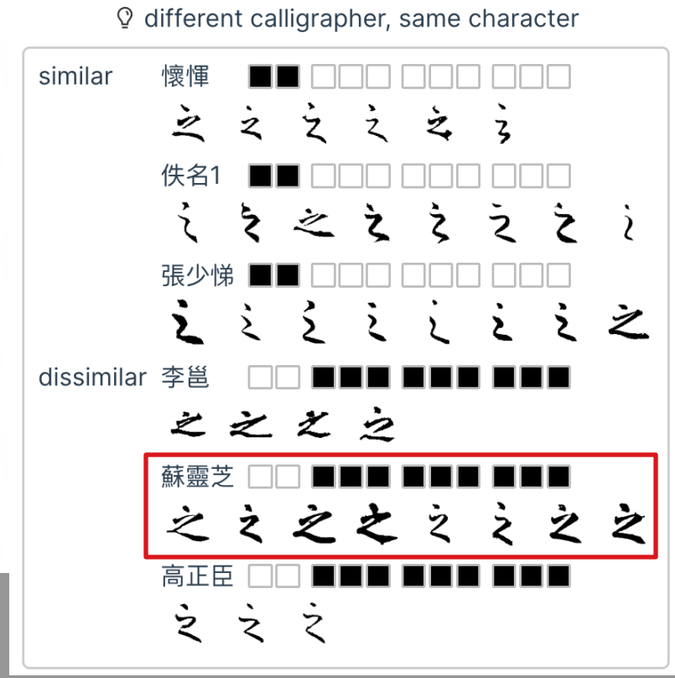
- Granularity Recommendation View: Wang is similar to Zhang Shaoti
  - Three clusters of Wang's *zhi* are largely preserved
  - Zhang's *zhi* falls within two of them
  - ⇒ Ming dynasty scholar Zhao Han: "Zhang deeply inherited the spirit of Wang's style"

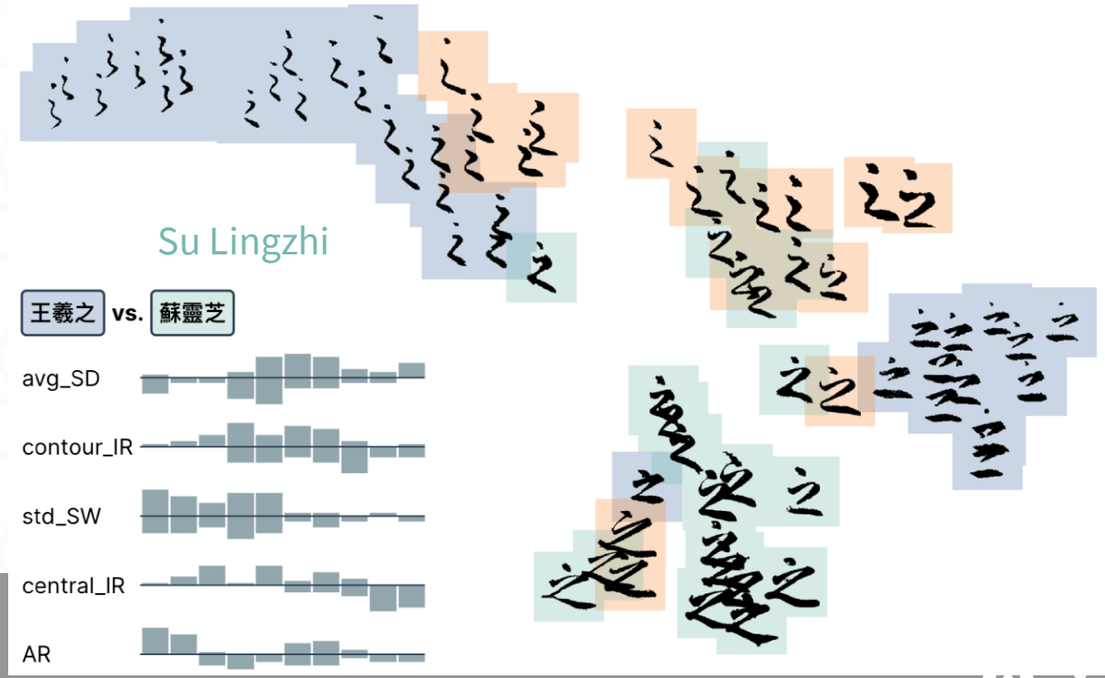




Case — Analyzing the Character *Zhi*

- Granularity Recommendation View: Wang is different from Su Lingzhi
  - Su's *zhi*: more compact (↓avg_SD, ↑central_IR, ↑contour_IR) and wider (↑AR)
  - ⇒ Emperor Xuanzong promoted a new aesthetic trend of fullness and plumpness
    - Su absorbed it into his own writing
  - ⇒ Song dynasty scholar Wei Tai: "Su's writing was so thick as to border on turbidity"





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Thanks!

Contact: jinchengli@bnu.edu.cn